8 With respect to an origin O, the position vectors of the points A and B are

$\overrightarrow{OA} = \begin{pmatrix} -3 \\ 20 \end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix} 6 \\ 8 \end{pmatrix}$.

(a) Determine whether $|\overrightarrow{AB}| > 200$. **[2]**

The point C is such that $\overrightarrow{AC} = \begin{pmatrix} 18 \\ -24 \end{pmatrix}$.

(b) Determine whether A, B and C are collinear. **[2]**

8(a)

8(b)